

2. Conditional Probability & Independence

Learning Outcomes

By the end of this week, you should be able to:

- Understand the definition, terminology, and uses of *conditional probability*
- Use *Bayes rule* to calculate conditional probabilities
- Understand *independence* in its various forms: *mutual, pairwise, conditional*
- Calculate *joint* probabilities using *independence* or the *multiplication rule*

Conditional Probability

- Imagine I roll a fair 6-sided die in the room next door
  - What is the probability of rolling 6?
  - If I told you the *roll was even*, what is the probability of rolling a 6?
- *Conditional probability* tells you how likely event A is to occur, *given knowledge that another event B has occurred*
  - Way of incorporating "partial information" into probability
- Definition: For events A, B, the conditional probability of A *given* B is denoted by $P(A \mid B)$ and calculated as

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

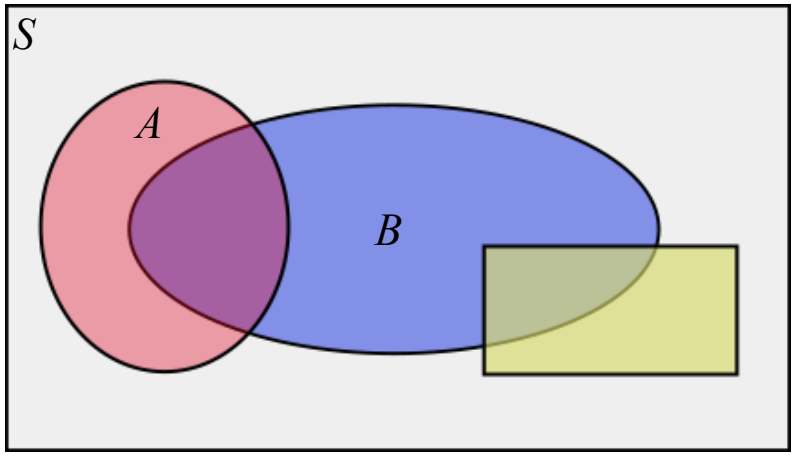
(provided $P(B) > 0$)

Conditional Probability

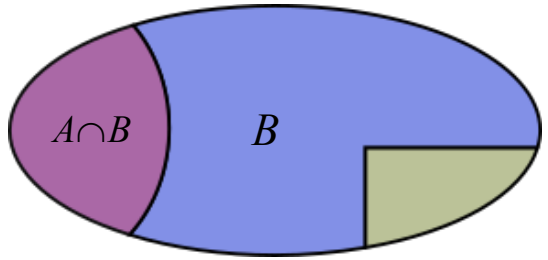
- Conditioning on an event B restricts sample space to B

- Conditional probabilities adjust original probabilities to restricted sample space B

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$



Unconditional space



Conditional on event B

Conditional Probability

- Conditional probabilities behave just like regular probabilities, i.e., they follow the probability axioms and all derived rules
  - $0 \le P(A \mid B) \le 1, \ \forall A \subseteq S$
  - $P(S \mid B) = 1$
  - If $A_1, A_2, A_3, \ldots$ are mutually exclusive then $P\left(\bigcup_{i=1}^{\infty} A_i \mid B\right) = \sum_{i=1}^{\infty} P(A_i \mid B)$
- Regular probabilities can be thought of as conditional on S

$$P(A) = P(A \mid S)$$

Example

- Consider rolling 2 dice, where:
  - A={“sum is even”}
  - B={“at least one 3”}
- Find conditional probs
  - $P(A \mid B)$
  - $P(B \mid A)$

A={“sum is even”}

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

B={“at least one 3”}

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

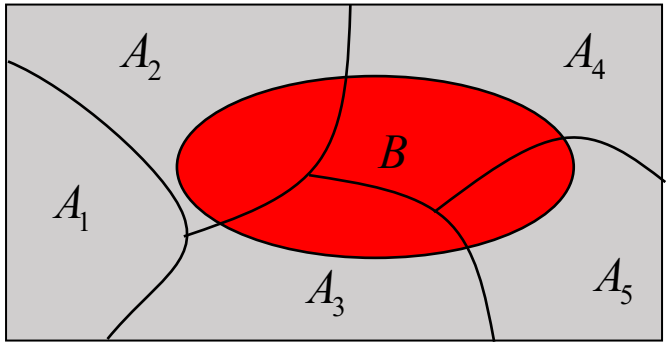
$A \cap B$

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

Example

- Consider probability space w/ events A, B s.t. $P(A), P(B)>0$
  - Find $P(A \mid B)$ when $A \subseteq B$
  - Find $P(A \mid B)$ when A, B are *disjoint*

Law of Total Probability



- Already know that for any partition $A_1, A_2, \ldots$

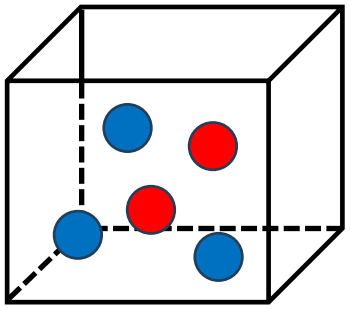
$$P(B) = P(B \cap A_1) + P(B \cap A_2) + \cdots$$

- Combined with $P(B \cap A) = P(B \mid A)P(A)$, we have

$$\boxed{P(B) = P(B \mid A_1)P(A_1) + P(B \mid A_2)P(A_2) + \cdots}$$

  - Calculate *unconditional* probs using partition & conditional probs
  - Simplest version $P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c)$

Example



- A box contains 3 Blue and 2 Red balls. Every time you pick a ball, you *put it back together with another ball of the same color*. What is the probability that the 2nd ball drawn is blue?

Bayes' Rule



- Assume you know $P(B \mid A)$, but you really want $P(A \mid B)$
  - E.g., know accuracy of screening test, but not whether you are sick
- *Bayes' rule* tells you how to swap conditioning events

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

  - Proof based on $P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A)$

Bayes Again

- For partition $A_1, A_2, \ldots$, Bayes' Rule is equivalently expressed as

$$P(A_j \mid B) = \frac{P(B \mid A_j)P(A_j)}{\sum_{i=1}^{n} P(B \mid A_i)P(A_i)}$$

  - Version to use depends on what is known ($P(B)$ vs $P(A_i)$'s)

  - Simplest case $P(A \mid B) = \dfrac{P(B \mid A)P(A)}{P(B \mid A)P(A) + P(B \mid A^c)P(A^c)}$

Example

- Doctor prescribes screening test for Hepatitis B (HB), as it affects 2% of population. However, test is not perfectly accurate!
  - If you have HB, test is positive (+) with $P(+ \mid HB) = .96$
  - If you are healthy, test is negative (−) with $P(- \mid HB^c) = .98$
  - Test's Confusion Matrix:

	Test Result	
Condition	**Pos**	**Neg**
Hep. B	**TP**	**FN**
Healthy	**FP**	**TN**

[T/F][P/N] = [True/False][Positive/Negative]

Example (cont'd)

- What is the *unconditional* probability of having HB?

- Find the *unconditional* probability of testing +

Example (cont'd)

- Find the *conditional* probability of HB, given that you tested +

- Find the *conditional* probability of HB, given that you tested −

Independence

- Two events A, B are called *independent* with respect to probability function/measure P when

$$\boxed{P(A \cap B) = P(A)P(B)}$$

  - For $P(A), P(B) \neq 0$, equivalent to $P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$

  - Independence *implies* that information on occurrence of one event *does not affect probability* of other

Example

- Can A, B be independent if they are *disjoint* ?

- Can A, B be independent if $A \subseteq B$?

Independence is *property of probability function,* not just events!

Example

- If A, B are independent (w.r.t. P) show that A, B^c are also independent

Mutual Independence

- Generalize independence to $n \geq 3$ events:

  Finite collection of events $A_1, A_2, \ldots, A_n$ is called (*mutually*) *independent* if for *any* sub-collection $A_{k_1}, A_{k_2}, \ldots, A_{k_m}$

$$P(A_{k_1} \cap A_{k_2} \cap \cdots \cap A_{k_m}) = P(A_{k_1})P(A_{k_2}) \cdots P(A_{k_m}) \Leftrightarrow$$

$$\Leftrightarrow P\left(\bigcap_{i=1}^{m} A_{k_i}\right) = \prod_{i=1}^{m} P\left(A_{k_i}\right)$$

- Definition based on *multiple* conditioning

Conditioning on Multiple Events

- For events A, B, C, conditional probability of A given B *and* C, denoted by $P(A|B,C)$, is

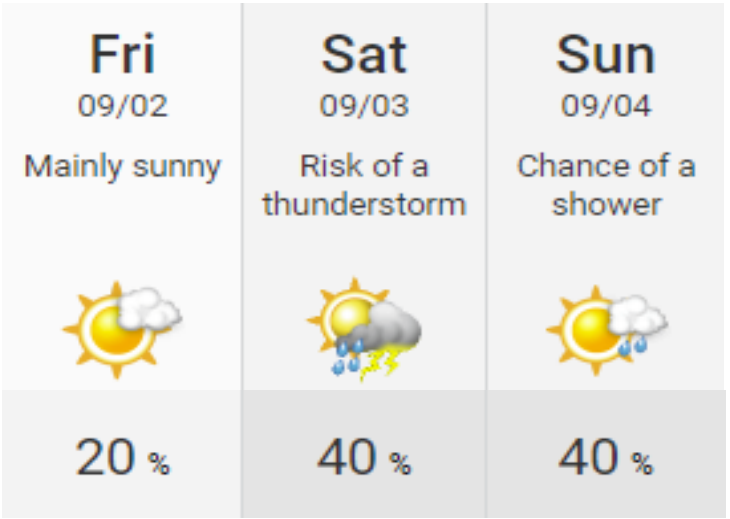
$$P(A \mid B,C) = P(A \mid B \cap C) = \frac{P(A \cap B \cap C)}{P(B \cap C)}, \text{ for } P(B \cap C) > 0$$

  - Extend similarly to ≥2 conditioning events

- If A, B, C, are *mutually independent*, then $P(A|B,C) = P(A)$ and similarly for $P(B|A,C) = P(B)$, $P(A \cap C \mid B) = P(A \cap C)$, etc
  - I.e., conditioning on *any* collection of events *does not affect probability* of intersection of any other events

Example

- Assume days are mutually independent; find probability it rains *only on one day.*



Mutual vs Pairwise Independence

- *Pairwise* independence *does not imply* mutual independence

  $P(A_i \cap A_j) = P(A_i)P(A_j),\ \forall i < j \not\Rightarrow A_1, A_2, \ldots, A_n$ mutually independent

- E.g. Consider flipping two coins & define events:
  $A=\{(H,H),(H,T)\}$, $B=\{(H,H),(T,H)\}$, $C=\{(H,H),(T,T)\}$

(H,H)	(H,T)
(T,H)	(T,T)

  - Are A, B, C pairwise independent?

  - Are A, B, C mutually independent?

Multiplication Rule of Probability

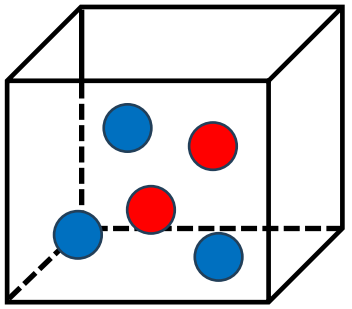
- Probability of *intersection* of $A_1, A_2, \ldots, A_n$ (a.k.a. *joint* probability) can be decomposed to product of conditional probabilities as

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) =$$
$$= P(A_1) \times P(A_2 \mid A_1) \times P(A_3 \mid A_2, A_1) \times \cdots \times P(A_n \mid A_{n-1}, A_{n-2}, \ldots, A_1)$$

  - Especially useful for *sequential* probability problems
  - For *mutually independent* events $A_1, A_2, \ldots, A_n$

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1)P(A_2)\cdots P(A_n)$$

Example



- A box contains 3 Blue and 2 Red balls. Every time you pick a ball, you put it back *together with another* ball of the same color. What is the probability that: 1st pick is red (R1), 2nd is blue (B2), and 3rd is red (R3)?

Conditional Independence

- Events A, B are called *conditionally independent* given C when

$$\boxed{P(A \cap B \mid C) = P(A \mid C)P(B \mid C)}$$

  - Implies that $P(A \mid B, C) = P(A \mid C)$ and $P(B \mid A, C) = P(B \mid C)$
  - Conditioning on 2 events is conditioning on their *intersection*

not ~~$P(A \mid B \mid C)$~~ but $P(A \mid B \cap C) = P(A \mid B, C)$

- Conditional independence *does not imply* independence, and vice-versa: $P(A \cap B \mid C) = P(A \mid C)P(B \mid C) \not\Rightarrow P(A \cap B) = P(A)P(B)$
  - *But* if A, B, C are mutually independent, they are also conditionally independent: $P(A \cap B \mid C) = P(A \cap B) = P(A)P(B) = P(A \mid C)P(B \mid C)$

Example

- Consider a fair (H/T) and a biased (H/H) coin. Pick one at random and flip twice; are the flip results independent?